



Islamic Specialized Bank — Low Risk (Community Banking)

Customer Risk Assessment Methodology

Customer Risk Rating / Customer Risk Assessment Model (CRA/CRAM)

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Document control

Document control history

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1.0	July 2026	Initial draft	

Approval sheet

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Related documents

Document	Reference
AML/CFT/CPF Compliance Policy	Version 1.0
Digital Onboarding and Due Diligence Policies & Procedures	Version 1.0
Sanctions & CPF Screening Procedures	Version 1.0
Transaction Monitoring Procedures	Version 1.0
End-to-End Customer Lifecycle and Onboarding SOP	Version 1.0
CRAM dataset workbook / factor-parameter map	Version 1.0

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1. Introduction

This document establishes the Customer Risk Assessment Methodology (“CRA/CRAM”) of Mal Bank LLC (“Mal” or “the Bank”), defining the policy, data, scoring, override, workflow and governance rules used to identify, assess, score, approve, monitor, reassess and evidence customer financial-crime risk at onboarding and throughout the customer lifecycle.

This Methodology is subordinate to, and operationalises, the Bank’s AML/CFT/CPF Compliance Policy (the “Policy”, Mal-CMP-AML-01) and must be read in conjunction with the Digital Onboarding and Due Diligence Policies & Procedures (Mal-PRC-ONB-01). Where any provision of this Methodology conflicts with the Policy, the Policy prevails and this Methodology shall be read down or amended accordingly. Where law, CBUAE requirements, sanctions obligations or formal supervisory direction impose stricter requirements, the stricter requirement prevails over this Methodology.

The Methodology establishes a single, internally consistent and CBUAE-aligned framework for calculating, approving and maintaining customer financial-crime risk ratings across enterprise and digital-bank customer populations. It is designed for implementation through a configurable rule engine, supported by controlled reference libraries, workflow routing, audit evidence, source-system lineage and model governance. The framework calculates an inherent customer risk score on a 1.0000 to 3.0000 scale, assigns a mathematical rating, applies hard stops and non-dilutable risk floors, and routes the customer to the required CDD, EDD, approval, monitoring and review outcome. Mathematical scoring is deliberately separated from mandatory overrides so that sanctions, PEP, STR/SAR, UBO verification failure, prohibited products, synthetic identity, failed digital identity and other critical risk indicators cannot be diluted by weighted averaging.

The primary risk domains covered are: money laundering (“ML”), terrorist financing (“TF”), proliferation financing (“PF”), targeted financial sanctions (“TFS”), sanctions evasion, high-risk country exposure, trade-based money laundering (“TBML”), fraud/scam/mule exposure relevant to financial crime, digital identity abuse and non-face-to-face channel risk (collectively “Financial Crime” or “FC”).

The authoritative model version for this Methodology is CRAM-CBUAE-2026-05-FREEZE-01 (approved model-freeze version for production build). This version is production-build ready from a methodology perspective; it is not, by itself, a production certification. Production deployment requires completion and independent review of calibration, source-system data mapping, vendor validation, SIT, UAT, parallel run, back-testing, evidence registration and governance sign-off (Section 17).

1.1 Objective

The objective of this Methodology is to enable consistent, explainable and auditable decisions on customer acceptance, CDD/EDD, approval authority, monitoring intensity, review frequency, restrictions, rejection, freezing, exit and regulatory escalation. Measurable objectives:

- Assign every in-scope customer and material related party a reproducible risk assessment record linked to a model_version_id;
- Calculate a raw mathematical score to at least four decimal places and apply rating thresholds to the unrounded raw score;
- Ensure 100% of mandatory factors and critical fields are either populated, blocked, or conservatively treated before approval;
- Apply hard stops and risk floors independently of the weighted score to prevent risk dilution;
- Generate reason codes and evidence references for all non-Low factors, overrides, data-quality exceptions and approval decisions;
- Trigger event-driven reassessment on material changes in customer profile, ownership, product, geography, digital channel, transaction behaviour, screening status or suspicion;
- Support independent validation, internal audit and regulatory reconstruction through point-in-time inputs, model versions, reference-library versions and immutable audit records.

1.2 Scope

This Methodology applies to all relevant employees, departments and functions involved in managing FC risks within the scope of their roles, including the First Line of Defence (“FLOD”), Compliance (Second Line of Defence, “SLOD”), and other control and support functions. It covers:

- New-to-bank onboarding, existing customer refresh, event-driven review, remediation, restriction, exit review and portfolio batch rescore;
- Natural persons including retail, payroll, affluent, HNW, joint accounts, minors, guardians, attorneys and authorised users;
- Legal persons including SMEs, free-zone entities, offshore entities, trusts/legal arrangements, SPVs, charities/NPOs, merchants and platform sellers;
- Financial institutions and regulated partners including correspondent/respondent relationships, PSPs, exchange houses, NBFIs, VASPs where permitted, fintech partners and payment facilitators;
- Digital channels including mobile, internet banking, digital onboarding, merchant portals, API/open-banking channels, remote servicing and high-risk remote account maintenance;
- Products including deposit accounts, salary accounts, cards, instant payments, domestic and cross-border transfers, merchant acquiring, payment gateway, virtual accounts/IBANs, wallets/stored value where offered, finance products, trade-related products and API/third-party access arrangements.

FATCA and CRS tax classification, self-certification, US indicia, tax residency, TIN, controlling-person and reasonableness logic are performed under the End-to-End Customer Lifecycle and Onboarding Operating Model (Mal-SOP-ONB-01) and are outside the scope of this Methodology. The tax classification outcome feeds this Methodology only as a customer-profile and geography risk input; this Methodology does not determine tax reportability. Sanctions and TFS screening, by contrast, are in scope of this Methodology as a mandatory pre-activation control and a determinant of hard stops and risk floors.

2. Governance & Oversight

The Bank establishes and maintains a robust governance and oversight framework to ensure the effective implementation, monitoring and continuous enhancement of the CRA/CRAM. This framework ensures that FC risks are appropriately identified, managed and controlled in line with applicable regulatory requirements and the Bank’s Board-approved Risk Appetite, and operates within the three-lines-of-defence model defined in Policy Section 9.

2.1 Roles & Responsibilities

Role	Responsibility
Senior Governance Forum / Board Risk & Compliance Committee (BRCC)	Approval forum for the Methodology and the frozen model version. Approves production go-live (Gate 6), threshold/weight changes, residual-risk acceptance, and receives model performance, KRI and portfolio-distribution reporting.
Model Governance Committee	Approves model freeze (Gate 0), threshold and weight changes, override-rule configuration and material model changes; monitors override volume/direction and model performance; ensures a single active approved parameter set per effective date.
Model Owner — Financial Crime Compliance / MLRO	Owns the Methodology and the model_version_id; maintains parameter libraries, floors, overrides and reason codes; approves reference-library updates; provides concurrence for High-risk outcomes per the Approval Matrix; decides STR/SAR-related floors and confidential handling.
Second Line of Defence — Compliance (Head of Compliance / MLRO)	Independent oversight and challenge of CRA outcomes; review of EDD adequacy; approval of manual downgrades (with documented rationale, expiry and evidence); sanctions/TFS escalation, freezing and regulatory reporting; sample reviews of Standard CDD outcomes.
First Line of Defence (FLOD)	Ensures completeness and accuracy of customer data feeding the model; validates that the system-generated rating is appropriate; executes EDD tasks and evidence collection; escalates inconsistencies, overrides and exceptions; maintains the expected-activity profile.
Data Governance	Owns source-to-target mapping, data lineage, data-quality testing and the exception register (Gate 1); maintains controlled, versioned reference libraries with maker-checker and effective-date control.
Vendor Governance / Fraud	Vendor due diligence and validation for screening, digital identity, device and fraud services (Gate 2); monitors vendor performance, drift and exception rates.

Role	Responsibility
Technology / Compliance Technology	Configures weights, thresholds, rules, libraries, workflows and reason codes through approved parameter tables; executes SIT (Gate 3) and deployment validation; maintains audit logging and security architecture.
Third Line of Defence — Internal Audit	Conducts periodic, risk-based independent assessments of the CRA/CRAM, including design and operating effectiveness and regulatory compliance; presents findings and validates closure of significant issues.

2.2 Core Design Principles

Principle	Mandatory implementation rule
Risk-based and evidence-led	Risk outcomes must be driven by documented risk indicators, controlled libraries, source-system data and approved judgement. Generic or unsupported risk assumptions are not permitted.
Non-dilution	Hard stops and mandatory floors override the mathematical score. A Low or Medium composite score must never suppress a prohibited or High-risk condition.
Inherent risk first	The primary rating is an inherent financial-crime risk rating. Residual controls may be used for management reporting, but must not reduce mandatory CDD/EDD treatment or override floors.
Point-in-time reproducibility	Every completed assessment must be reproducible from the customer data, reference libraries, model version, rule version and approvals effective at the time of assessment.
No default-to-Low	Missing, unmapped, inconsistent or pending critical data must block completion or be conservatively treated. No field may default to UAE, Low, “none” or zero unless sourced and evidenced.
Configurable governance	Weights, thresholds, rules, libraries, workflows and reason codes must be configurable through approved parameter tables, not hard-coded, except where security architecture requires controlled code deployment.
Lifecycle sensitivity	New customers rely more on profile, product, geography and digital onboarding assurance. Existing customers must place higher weight on actual behaviour and emerging risk events.
Auditability and explainability	Users must be able to see factor scores, parameter scores, top drivers, overrides, workflow status, evidence and next review date.

2.3 Regulatory Alignment and Model Interpretation Rules

This Methodology is aligned with Federal Decree-Law No. (10) of 2025 and Cabinet Resolution No. (134) of 2025 (the Executive Regulations) under the UAE AML/CFT/CPF legal framework, Cabinet Resolution No. (74) of 2020 on Targeted Financial Sanctions, and applicable CBUAE rules, guidelines, notices and circulars.

Regulatory expectation	Methodology response
Risk-based CDD and EDD	Customer risk assessment is performed at onboarding, refresh and event review. Ratings drive CDD, EDD, approval authority, monitoring intensity and review frequency.
Targeted Financial Sanctions	UN/UAE TFS and sanctions screening must be completed before activation and on list updates. Confirmed true matches trigger hard stop, freeze/escalation/reporting as required.
Non-face-to-face risk mitigation	Digital onboarding/channel assurance is a standalone factor with identity, document, biometric, liveness, device, geolocation, authentication and link-analysis controls.
Ongoing monitoring	Expected vs actual behaviour, rolling transaction windows, velocity, pass-through, counterparty concentration, alert outcomes and STR/SAR events trigger reassessment.
CDD/KYC and recordkeeping	All inputs, transformations, scores, overrides, approvals, evidence and model/library versions must be retained in an immutable assessment record.
Independent validation	The Methodology, configuration, calibration, data mapping, vendor outputs, SIT, UAT, parallel run and back-testing must be independently validated before production go-live and periodically thereafter.
Regulatory precedence	Where law, CBUAE requirements, sanctions obligations or formal supervisory direction impose stricter requirements, the stricter requirement prevails over this Methodology.

3. Authoritative Scoring Framework

3.1 Score Scale and Rating Bands

Raw score band	Mathematical rating	Minimum treatment	Implementation rule
1.0000–1.5000	Low	Standard CDD; SDD only where legally permitted and explicitly evidenced (Policy 13.1; Mal-PRC-ONB-01 Section 7.4).	Apply only where no override, floor, missing critical data or unresolved screening issue exists.
1.5001–2.1500	Medium	Standard CDD plus targeted controls, clarifications and monitoring adjustments.	Medium floor rules may force this rating even where the raw score is Low.
2.1501–3.0000	High	EDD, enhanced monitoring, senior/delegated approval and shorter review cycle.	High also applies where a mandatory floor or override is triggered.
Override only	Prohibited / Reject / Freeze / Exit	No relationship, or restricted legal/sanctions handling as required.	Cannot be downgraded by business user or mathematical score.

Authoritative threshold decision. The only approved Low/Medium boundary is 1.5000/1.5001. Any prior Low/Medium boundary different from 1.5000/1.5001 is retired and must not appear in configuration, UAT scripts, MI reports, calibration packs or user documentation.

3.2 Mathematical Formula and Calculation Steps

For each assessment, the model performs the following in sequence:

1. Identify customer segment, lifecycle stage and material related-party roles;
2. Load the active approved model_version_id, factor weights, parameter weights, thresholds, libraries and override rules;
3. Validate mandatory data completeness and screening prerequisites;
4. Score parameters using controlled libraries and deterministic rules;
5. Calculate factor scores by weighted average, maximum-risk rule or approved range logic, as configured;
6. Calculate raw composite score = $\text{sum}(\text{factor_score} \times \text{factor_weight})$;
7. Map the raw composite score to Low, Medium or High using the authoritative rating bands;
8. Apply hard prohibitions, High floors and Medium floors in priority order;
9. Produce final rating, CDD/EDD outcome, approval route, monitoring profile and next review date;
10. Persist the full audit record and publish the approved rating to source and downstream systems.

3.3 Rounding and Precision Controls

- Raw composite scores must be stored to at least four decimal places; six decimal places are preferred for configuration validation;
- Threshold decisions must use the unrounded raw score. Display rounding must never drive the final rating;
- All weights must be stored as decimals, not only displayed percentages (e.g., 16.666667% is stored as 0.166667);
- Deployment weight-total tolerance is +/- 0.0001 per factor set. Any active segment/lifecycle weight set outside tolerance must fail deployment;
- No intermediate contribution may be rounded before summation.

4. Customer Segments and Relationship Hierarchy

Code	Segment	Roles captured	Risk treatment
NP	Natural Person	Customer, joint holder, minor, guardian, attorney, authorised user, cardholder, sole establishment owner where applicable.	Use NP profile, geography, product, digital-channel and expected/actual activity factors. Apply the highest material related-party risk where the party can transact, control or benefit.

Code	Segment	Roles captured	Risk treatment
LP	Legal Person / SME	Company, partnership, trust, foundation, SPV, NPO/charity, UBO, shareholder, director, manager, signatory, POA, controller, parent.	Use entity profile, ownership/control, business activity, geography, product, channel and activity factors. Unverified UBO or no lawful purpose is a hard stop or High pending remediation.
MER	Merchant / Platform Seller	Merchant entity, beneficial owner, administrator, website/app owner, settlement account, payment facilitator or platform relationship.	Use LP scoring plus merchant MCC, chargeback/refund risk, card-not-present exposure, payout velocity, settlement and digital presence rules.
FI	Financial Institution / Regulated Partner	Respondent, correspondent-style relationship, PSP, exchange house, NBFI, VASP where permitted, fintech partner, payment facilitator, downstream/nested exposure.	Use the FI six-factor taxonomy including standalone systems and controls. Correspondent, nested, downstream or high-risk corridor exposure requires EDD and senior approval.
EMP	Employee / High-Risk Insider Relationship	Staff customer, staff-related business interest, privileged user with customer relationship exposure.	Track as a relationship risk indicator and conflict/insider risk overlay. Do not reduce or replace customer CRR scoring.

5. Factor Weights by Segment and Lifecycle Stage

5.1 Active Production Factor Weights

Factor	NP New	NP Existing	LP/MER New	LP/MER Existing	FI New	FI Existing
Customer profile risk	25%	20%	25%	20%	20%	15%
Geography / sanctions nexus risk	20%	15%	20%	15%	20%	20%
Systems and controls	N/A	N/A	N/A	N/A	15%	15%
Product / service / correspondent risk	20%	15%	20%	15%	20%	20%
Digital onboarding / channel / API assurance	25%	20%	20%	15%	10%	10%
Expected activity / transaction behaviour / funding	10%	30%	15%	35%	15%	20%
Total	100%	100%	100%	100%	100%	100%

FI taxonomy freeze decision. The six-factor FI model is the only approved FI / Regulated Partner taxonomy. Any FI configuration that does not include a standalone Systems and Controls factor is retired and must fail configuration validation.

5.2 Legacy and Traditional Compatibility

Where a traditional non-digital onboarding channel or legacy customer population is migrated into the model, the customer must still be assigned to the active segment/lifecycle factor set above. Legacy split geography fields may be mapped into the active geography factor but must not create a second active model. If an individual geography split is retained for reporting, residence must be stored as 0.166667 and nationality as 0.083333 within the 0.250000 total geography allocation; the deployment gate must validate the full-factor total, not displayed percentage strings.

6. Parameter Libraries and Scoring Rules

Scoring is two-layer: the parameters in this Section roll up into the composite factors, which combine at the segment and lifecycle weights in Section 5.1. Every non-Low parameter requires a reason code and source field. All parameter weight sets must reconcile to 100% within deployment tolerance.

6.1 Natural Person Profile Parameter Library

Parameter	Weight	Low = 1	Medium = 2	High = 3	Required control
Employment / occupation risk	15%	Salaried employee, pensioner or student with verified sponsor/income source.	Self-employed, freelancer, commission-based, variable income or occupation pending clarification.	Unemployed with high activity, high-risk occupation, cash-intensive occupation or unclear source of funds.	Occupation library linked to source-of-funds evidence rules.
Customer segment	10%	Mass retail/payroll with verified employer and ordinary product needs.	Affluent, SME owner, freelancer, non-payroll or multiple product needs.	HNW, private-banking-like profile, politically connected exposure or high-value relationship.	Segment derived from product eligibility and customer profile.
PEP exposure	15%	No PEP exposure.	Domestic PEP, former PEP or low-materiality associate where policy permits Medium treatment.	Foreign PEP, IO PEP, close associate/family of material PEP or high-risk PEP nexus.	PEP screening and declaration reconciliation; High floor where applicable.
Source of funds quality	15%	Verified salary, pension or regulated employer income.	Self-declared income with reasonable evidence or partially verified business income.	Third-party funds, unexplained funds, complex or inconsistent source.	Mandatory evidence by risk level.
Source of wealth complexity	10%	Not applicable or simple verified accumulation.	Business ownership, investments or inheritance with partial support.	Complex offshore wealth, crypto-derived wealth or unverified wealth.	EDD required where High.
Residency / legal status	10%	Resident with verified address and legal status.	Non-resident in acceptable country with strong evidence, or short-term residence with explanation.	Unclear residence, conflicting addresses, transient profile or unable to verify.	Non-resident may create Medium or High floor depending on evidence.
Adverse intelligence	15%	No material adverse media or internal negative history.	Low/moderate adverse media with mitigation.	Material adverse media, prior exit, law-enforcement concern or financial-crime allegation.	Material true adverse media is High floor or reject.
Customer cooperation / transparency	10%	Complete, consistent and timely information.	Minor gaps resolved before activation.	Refusal, inconsistent explanation or inability to verify identity/source.	Failure to complete CDD is reject/exit.

6.2 Legal Person / SME / Merchant Profile Parameter Library

Parameter	Weight	Low = 1	Medium = 2	High = 3	Implementation rule
Legal form and structure	15%	Simple UAE entity, listed/publicly supervised entity or transparent operating company.	Standard LLC/SME, free-zone entity or ordinary ownership structure.	Trust/foundation/nominee/bearer-like, offshore, SPV, multi-layer or opaque structure.	Complex structure floors at least Medium; opacity floors High or reject.
Ownership and control transparency	20%	UBOs and controllers identified and verified; simple ownership.	Multiple layers but UBO/control verified.	UBO unclear/refused, nominee concerns, circular ownership or inability to verify.	Refusal or inability to identify UBO is reject or High

Parameter	Weight	Low = 1	Medium = 2	High = 3	Implementation rule
					pending remediation.
Business activity / industry	20%	Low-risk professional, retail or ordinary service activities.	General trading, e-commerce, import/export or medium-risk industry.	DPMS, arms/defence, unlicensed financial activity, NPO/charity high risk, high-risk sectors or virtual asset nexus.	Industry library maintained and mapped to ISIC/MCC.
Operating history	7.5%	Operating >3 years with evidence.	Operating 1–3 years or limited evidence.	Newly incorporated with high expected turnover or unusual activity.	New entity not prohibited but may trigger Medium/High floor.
Regulatory / licensing status	10%	Licence verified and activity aligned.	Licence pending renewal or minor mismatch remediated before activation.	Unlicensed regulated activity, expired/suspicious licence or activity outside licence.	Unlicensed regulated activity is prohibited/reject.
Related-party PEP/sanctions/adverse exposure	12.5%	None.	Domestic PEP or moderate adverse intelligence with mitigation.	Foreign PEP, material adverse intelligence, sanctions nexus or high-risk controller.	Highest material related-party risk floors the customer rating.
Business model transparency	10%	Clear customers, suppliers, revenue model and economic purpose.	Partially explained but plausible business model.	No apparent lawful economic purpose, inconsistent model or shell indicators.	No lawful purpose is reject/exit.
Merchant / digital presence	5%	Verified website/store, consistent MCC and transparent settlement.	New website/marketplace, moderate chargeback/refund exposure.	Hidden ownership, deceptive activity, high-risk digital goods, high refund/chargeback exposure.	Merchant risks feed product and behaviour factors.

6.3 Financial Institution / Regulated Partner Parameter Libraries

6.3.1 FI Customer Profile Factor Parameters

Parameter	Weight	Low = 1	Medium = 2	High = 3	Evidence
Nature of FI activity	25%	Standard regulated banking/financial activity with no high-risk services.	Regulated FI with material cross-border, wholesale or higher-risk services within appetite.	Correspondent, nested/downstream, payable-through, MSB/exchange, stored value, VASP, private banking or trade-finance high-risk activity.	Licence, questionnaire, product/service schedule.
Legal structure and physical presence	20%	Simple regulated legal structure with verified physical presence.	Group structure with added complexity but transparent and verified.	Shell bank, online-only bank without adequate licensing/physical presence, unlicensed entity or structure outside appetite.	Corporate registry, licence, physical-presence evidence.
Ownership and control transparency	20%	Transparent ownership/control with no unexplained recent changes.	Complex ownership or recent ownership change but verified and explained.	Bearer/nominee/trust opacity, unidentified UBO or inability to verify control persons.	Ownership chart and UBO evidence.
Regulatory oversight	20%	Regulator applies FATF standards; no material AML/CFT/CPF concern.	Regime or enforcement history requires enhanced review.	Weak regulator, FATF high-risk jurisdiction, unclear licence or serious AML/CFT/CPF enforcement action.	Regulator review and enforcement screening.

Parameter	Weight	Low = 1	Medium = 2	High = 3	Evidence
Adverse regulatory / financial-crime history	15%	No material enforcement, sanctions, fraud or adverse media history.	Minor or remediated issue with evidence.	Recent serious AML/CFT/CPF sanction, sanctions evasion, fraud, corruption or unresolved adverse intelligence.	Adverse media, regulatory notices, audit reports.

6.3.2 FI Systems and Controls Factor Parameters

The FI systems-and-controls factor enters the composite once, at its Section 5.1 factor weight. FI control parameters roll up into this factor and are not added to the composite independently.

Parameter	Weight	Low = 1	Medium = 2	High = 3	Evidence
KYC/CDD and sanctions policy	20%	Written, approved and recently updated AML/CFT/CPF/TFS policies.	Policy exists but is outdated, incomplete or weakly evidenced.	No adequate policy evidence or refusal to provide.	Policy pack, approval date, gap assessment.
TFS/sanctions screening capability	20%	Automated screening of customers, related parties and transactions with list updates and escalation.	Partial/manual screening or limitations requiring compensating controls.	No effective screening evidence or unresolved control weakness.	Screening architecture, list sources, SLAs, sample alerts.
AML monitoring and STR/SAR framework	20%	Documented monitoring, escalation and reporting framework with governance evidence.	Framework exists but gaps or limited evidence remain.	No documented framework or material weakness.	TM coverage, SAR/STR procedures, governance minutes.
PEP/adverse media controls	15%	Robust PEP/adverse media screening and periodic refresh.	Partial controls or manual reliance.	No effective PEP/adverse media control evidence.	Screening results, procedures, QA evidence.
Correspondent/nested customer controls	15%	Clear downstream customer prohibition or controls, payable-through restrictions and respondent due diligence.	Limited downstream visibility but mitigants documented.	Unknown downstream exposure, nested relationships or payable-through features without control.	Questionnaire, contractual controls, respondent assessment.
Independent audit, training and governance	10%	Recent independent AML audit, remediation tracking, role-based training and governance reporting.	Audit/training exists but limited or stale.	No independent review, training or governance evidence.	Audit report, training logs, issue register.

7. Geography and Country Risk Methodology

7.1 Country Risk Library

Country risk is expressed through the following pillars, each carrying a single coherent value domain. The composite country risk level is derived from the weighted pillars. Country reference libraries are maintained as controlled, versioned, executable data with maker-checker and effective-date control.

Country risk pillar	Weight	Low = 1	Medium = 2	High = 3	Mandatory rule
Sanctions / TFS exposure	30%	No relevant sanctions nexus.	Targeted or sectoral sanctions exposure requiring monitoring.	Comprehensive sanctions, UN/UAE TFS exposure or prohibited nexus.	Confirmed UN/UAE TFS true match = Prohibited / freeze / escalate.
FATF status	20%	Not subject to increased monitoring or call for action.	Increased monitoring or grey-list equivalent.	Call for action, countermeasure or equivalent.	Follow current regulatory requirements and risk appetite.

Country risk pillar	Weight	Low = 1	Medium = 2	High = 3	Mandatory rule
PF, terrorism, conflict and instability	10%	No material PF/terrorism/conflict concern.	Moderate concern or regional exposure.	High PF, terrorism, conflict or evasion risk.	PF high-risk nexus floors the relevant customer/product to High.
Corruption and transparency	10%	Strong public integrity and transparency.	Moderate corruption/secrecy concerns.	High corruption, secrecy or weak BO transparency.	High corruption plus complex structure floors the entity to High.
AML/CFT supervisory quality	10%	Strong supervision and enforcement.	Moderate supervisory weakness.	Strategic deficiencies or weak enforcement.	Applies materially to FI/partner due diligence.
Predicate offence exposure	5%	Low known exposure.	Moderate organised crime/narcotics/predicate exposure.	High exposure to organised crime/narcotics/major predicates.	High exposure increases EDD and monitoring.
Tax transparency / offshore risk	5%	Transparent jurisdiction.	Moderate secrecy/offshore features.	High secrecy, nominee or opaque corporate services.	Relevant to UBO/SOW complexity.
Internal typology and bank exposure	10%	No significant internal typology findings.	Moderate alerts/losses/cases.	High STR/SAR, fraud, mule, sanctions or exit experience.	Compliance may escalate a country rating based on internal evidence.

7.2 Customer-Specific Geography Attributes

Customer type	Attribute	Weight within geography factor	Scoring method	Floor / treatment
NP	Residence country	25%	Lookup country library rating.	High country residence floors to High unless prohibited.
NP	Primary/secondary nationality	20%	Highest score across nationalities.	Sanctions/TFS nationality nexus triggers screening escalation.
NP	Country of birth	10%	Lookup where available; capture unavailable reason.	Does not alone floor to High unless combined with other indicators.
NP	SOF/SOW country	15%	Highest country score across recurring funds and wealth origin.	High-risk source country requires EDD evidence.
NP	Expected transaction corridors	15%	Highest corridor score or exposure-weighted method.	High-risk corridor enhances monitoring.
NP	Digital geolocation signals	15%	Highest risk among IP/GPS/SIM/device inconsistency.	VPN/TOR/high-risk mismatch may trigger Medium/High floor.
LP/MER	Incorporation, operation, UBO/controller and settlement countries	Weighted / highest-risk mix	Country library lookup and related-party materiality.	High-risk or prohibited country exposure floors Medium/High/Reject based on risk appetite.
FI	Regulator, licensing, ownership, downstream exposure and corridors	Weighted / highest-risk mix	Country library plus regulator-quality assessment.	Weak regulator, unknown downstream exposure or high-risk corridor floors High.

Jurisdiction-based restrictions are driven primarily by official sanctions and TFS sources — the UN Security Council Consolidated List, the UAE Local Terrorist List and EOCN notifications — supplemented by the Bank's approved country-risk methodology (Mal-PRC-ONB-01 Section 7.3; Policy Sections 13.6 and 16). Real-time screening against the current official lists takes precedence over any static list.

8. Product and Service Risk Methodology

8.1 Product Risk Driver Library

Driver	Weight	Low = 1	Medium = 2	High = 3	Implementation rule
Liquidity and convertibility	10%	Low liquidity or restricted usage.	Moderate liquidity.	Instantly liquid, transferable or cash-equivalent.	Cards, wallets and instant payments typically score higher.
Velocity / speed	10%	Slow settlement or controlled disbursement.	Same-day or limited rapid movement.	Real-time/high-velocity movement.	Higher velocity increases layering and mule risk.
Cross-border capability	10%	Domestic only.	Limited corridors/currencies.	Open or high-risk cross-border corridors.	High-risk corridor may trigger product/geography High.
Third-party payment capability	10%	Own-account only.	Limited third-party transfers.	Broad third-party/beneficiary payments.	Requires beneficiary screening and monitoring.
Anonymity / opacity	10%	Transparent named customer only.	Some intermediary opacity.	Obscured ownership, virtual identifiers or nested users.	Anonymous or bearer-like product is prohibited.
Value scalability / limits	8%	Low limits and low balances.	Moderate limits.	High value, scalable or rapidly changeable limits.	Limit increases trigger pre-activation rescore.
Cash/cash-equivalent exposure	8%	No cash-equivalent exposure.	Limited exposure.	Cash, prepaid, crypto/VA, high cash-equivalent functionality.	Cash-equivalent exposure floors Medium/High.
Trade/PF/dual-use exposure	8%	No trade/PF relevance.	Limited goods/counterparty exposure.	Trade finance, dual-use goods or high-risk shipping/corridors.	PF indicators trigger High.
Fraud/scam/mule exposure	10%	Low susceptibility.	Moderate susceptibility.	Known mule/scam/ATO exploitation or high fraud typology.	Fraud and AML signals must be linked.
Merchant/card-not-present exposure	6%	Not applicable or low-risk card-present.	Standard e-commerce merchant risk.	High chargeback/refund, digital goods or deceptive model.	Merchant acquiring requires a merchant score.
API/open banking/automation exposure	5%	No third-party/API access.	Controlled API with limited scope.	Bulk/API/TPP access enabling nested activity.	Partner/API due diligence required.
Control weakness indicator	5%	No material control weakness.	Known limitation with remediation plan.	Weak, untested or failed product controls.	Strong controls cannot reduce inherent score; weak controls may increase risk or require remediation.

8.2 Product Risk Treatment Rules

- Each product must have an approved product risk score before launch or customer activation;
- Where a customer has multiple products, the default customer product factor uses the highest active or requested product score unless a validated exposure-weighted method is approved;
- Product features such as cross-border enablement, API access, merchant activation, virtual IBANs, limit increases or high-risk corridors must trigger pre-activation reassessment;

- Strong controls may prevent additional uplift or support risk acceptance but must not reduce inherent product risk below the risk implied by product features;
- Weak controls must either increase the product risk score, trigger remediation, restrict activation or require risk acceptance depending on severity.

8.3 Digital-Bank Product Baseline Matrix

Product / service	Baseline risk	Rationale	System treatment
Basic current/savings account	Medium	Entry account and potential mule account depending on funding and transfer capability.	Medium unless restricted to verified salary/own-account transfers and low limits.
Salary/payroll account	Low–Medium	Lower risk where employer and payroll source are verified.	Medium where employer is high risk or source unclear.
Debit card	Medium	Cash withdrawals, online purchases and card-not-present exposure.	High where cross-border/high-risk MCC/crypto/gambling exposure is enabled without controls.
Credit or virtual card	Medium–High	Cash advances, high velocity, online use, chargeback and scam typologies.	High for high limits or instant virtual issuance with weak identity assurance.
Domestic transfers / instant payments	Medium–High	Rapid movement and mule/scam risk.	High for high limits, new-beneficiary velocity or weak authentication.
International transfers / remittances	High	Cross-border, sanctions, corridor, third-party and layering exposure.	High by default; corridor controls and beneficiary screening mandatory.
Wallet / stored value	High	Prepaid/cash-equivalent, rapid movement and possible third-party use.	High by default; prohibited if anonymous or unverified.
Virtual accounts / virtual IBANs	High	Opacity, reconciliation complexity, nested customer and third-party payer risk.	High; requires LP/MER/FI due diligence and monitoring.
Merchant acquiring / payment gateway	High	Merchant fraud, card-not-present laundering, chargebacks and refund abuse.	High for high-risk MCCs, digital goods, cross-border merchants or payment facilitators.
Open banking / API access	Medium–High	Third-party access, automation and downstream control exposure.	High for bulk payment initiation, partner aggregation or nested customers.
Trade finance / invoice finance	High	TBML, sanctions, dual-use goods, over/under-invoicing and shell-company risk.	High by default; PF and goods screening required.
Crypto/virtual asset exposure where not licensed	Prohibited / Restricted	Requires specific licensing, risk appetite and controls.	Reject/block unless expressly permitted by licence, policy and controls.

9. Digital Onboarding and Channel Assurance

In recognition of Digital Onboarding and Non-Face-To-Face (“NF2F”) relationships, digital onboarding/channel assurance is a standalone risk factor. It operates alongside the digital onboarding control framework and Digital Onboarding Risk Matrix set out in Mal-PRC-ONB-01 (Sections 5, 6 and Appendix 4) and the Digital ID Assurance and Appropriateness Assessments required under Policy Section 12.3.1.

9.1 New Customer Digital Onboarding Parameters

Parameter	Weight	Low = 1	Medium = 2	High = 3	Floor / control
Identity proofing source	20%	UAE Pass/Emirates ID or approved reliable independent source.	Approved IDSP with documentary and database checks.	Document upload only, weak or unknown source.	Weak source floors High or onboarding hold.
Document authenticity	12.5%	Automated authenticity pass and no tampering.	Manual review or partial validation resolved.	Fail, tampering, expired/invalid document.	Fail = reject/remediate before scoring.

Parameter	Weight	Low = 1	Medium = 2	High = 3	Floor / control
Biometric match and liveness	17.5%	Strong biometric match and liveness pass.	Inconclusive but remediated with approved method.	Mismatch, liveness fail, suspected deepfake/injection.	High floor; confirmed fraud = reject.
Device integrity	10%	Normal device; no emulator/root/jailbreak.	Unknown or new device with limited history.	Rooted/jailbroken, emulator, tampered, device farm.	High floor where combined with weak ID or high-risk geography.
IP/GPS/SIM/geolocation consistency	10%	Consistent with declared profile.	Minor mismatch explained.	High-risk, impossible travel, spoofed or anonymised location.	Compliance/fraud review.
Application velocity / link analysis	10%	No duplicate or suspicious links.	Moderate duplicate/household links.	Multiple applications from same device/IP/biometric/contact data.	Fraud review and possible rejection.
Authentication and credential binding	10%	Device binding plus biometric/passkey/MFA.	App OTP/SMS OTP with other controls.	Password/SMS-only for high-risk actions.	Step-up before activation.
Vendor/model assurance	5%	Vendor independently assured and monitored.	Vendor approved but limited performance history.	Vendor untested or material issue.	Cannot rely without risk acceptance.
Exception/fallback handling	5%	No exception or approved evidence-based fallback.	Manual exception with documented approval.	Repeated or undocumented overrides.	Compliance approval required.

9.2 Existing Customer Digital Channel Parameters

Parameter	Weight	Low = 1	Medium = 2	High = 3	Trigger action
Authentication strength	20%	Biometric/passkey/MFA with device binding.	MFA present but not phishing resistant.	Weak authentication or SMS-only for high-risk events.	Step-up and channel score increase.
Device change behaviour	15%	Stable device pattern.	Normal new device with verification.	Frequent device changes, emulator/root or device farm.	Hold high-risk transactions.
Credential lifecycle events	15%	Normal password/MFA management.	Occasional reset/change.	SIM swap, email change + new device + beneficiary setup.	ATO/scam review and rescore.
IP/geolocation anomaly	15%	Consistent location.	Moderate inconsistency.	Impossible travel, high-risk country, VPN/TOR for high-risk event.	Review; Medium/High floor.
High-risk digital actions	15%	No high-risk action or normal action.	Beneficiary/limit/contact change with controls.	Stacked actions: new beneficiary + limit increase + large transfer.	Hold/step-up/recalculate.
API/open banking access	10%	No API/TPP access or low scope.	Limited-scope TPP access.	Bulk payments, unattended API or partner access.	Partner/API controls required.
Fraud/scam/mule indicators	10%	No indicators.	Low/moderate resolved indicator.	Confirmed or repeated mule/scam/ATO indicators.	High floor and case management.

10. Expected Activity, Behaviour and Funding Risk

Expected activity is collected at onboarding and validated after activation (Mal-PRC-ONB-01 Sections 8.4 and 9.4; Policy Section 12.5). Existing-customer scoring must use observed behaviour in rolling windows. Behavioural risk is not limited to monetary thresholds: the model must also detect velocity, pass-through, counterparty concentration, unexpected corridors, dormancy-to-activity, funding-source anomalies and alert/case outcomes.

Behavioural metric	Medium trigger	High trigger	System action
Turnover deviation	Actual monthly turnover 2x expected without explanation.	Actual monthly turnover >3x expected, high-risk corridor involved, or activity inconsistent with profile.	Create review task; High trigger recalculates immediately.
New-beneficiary velocity	3–5 new beneficiaries in 7 days.	More than 5 new beneficiaries in 7 days or large transfer to new beneficiaries.	Step-up authentication and TM scenario.
Pass-through ratio	>60% of funds leave within 48 hours without explanation.	>80% of funds leave within 24 hours with multiple counterparties.	Mule/layering review and possible High floor.
Third-party funding	Material third-party funds inconsistent with profile.	Repeated third-party funding from unrelated/high-risk parties.	EDD, restrictions or exit review.
Cross-border exposure	Unexpected corridor appears.	High-risk corridor or sanctions/high-risk adjacency appears.	Rescore geography/product/transaction factor.
Dormancy then sudden activity	Dormant account resumes material activity.	Dormant account resumes high-volume/high-risk corridor activity.	Event review and possible restriction.
Rejected/fraudulent card or merchant activity	Elevated but explainable rejection/chargeback/refund pattern.	High chargeback/refund/fraud rate or deceptive merchant indicator.	Merchant/customer rescore and potential restriction.
STR/SAR or confirmed suspicion	Any confirmed suspicion outcome.	STR/SAR filed or suspicion cannot be mitigated.	High floor; confidential handling and tipping-off control (Mal-PRC-ONB-01 Section 16).

11. Screening, Sanctions, PEP and Adverse Intelligence

Screening is performed in accordance with the Sanctions & CPF Screening Procedures and Mal-PRC-ONB-01 Section 11. Sanctions/TFS screening is a mandatory hard-stop control; no onboarding proceeds without clearance, and confirmed matches are frozen without delay and reported (FFR/PNMR via goAML) per Policy Section 16.5.

Screening type	When performed	System output	Risk outcome
UN/UAE TFS and sanctions	Pre-onboarding, ongoing list updates, before material activation/transactions as required.	Clear, potential match, true match, false positive, pending.	True match = hard stop/freeze/escalate/report; pending = hold.
PEP screening	Pre-onboarding and ongoing refresh for customers and related parties.	PEP category, role, country, relationship and date.	Foreign/IO PEP usually High; domestic PEP Medium/High based on role and controls.
Adverse media	Pre-onboarding, periodic and trigger-based.	Severity, category, source, date, disposition.	Material true adverse media floors High or reject.
Internal watchlist / exited customer	Pre-onboarding and event review.	Prior exit, rejected onboarding, fraud/mule, suspected sanctions evasion.	Financial-crime exit floors High or reject.
Law enforcement / regulatory inquiry	Event-based.	Inquiry category, status and evidence.	Review trigger; High where material.
VASP/crypto exposure screening	Where applicable by policy and product.	Direct/indirect exposure, wallet/VA linkage where available.	High or prohibited based on licence and risk appetite.

12. Overrides, Risk Floors and Prohibitions

12.1 Override Hierarchy

Overrides are deterministic, auditable and separate from the weighted scoring engine. The system stores all applicable overrides but applies the most restrictive final outcome. Hard prohibitions are evaluated before normal approval

routing; High and Medium floors are then applied after the mathematical score. Manual downgrades cannot override hard prohibitions or mandatory floors. Prohibited customer types and acceptance boundaries are defined in Mal-PRC-ONB-01 Section 7 and Policy Sections 8.1–8.2.

Rule ID	Trigger	Outcome	Priority	Required evidence / treatment
OVR-001	Confirmed UN/UAE TFS or sanctions true match, legally required freeze or prohibited sanctions exposure.	Hard stop / Freeze / Report	Highest	Sanctions case record, disposition, freeze/escalation/reporting evidence.
OVR-002	Prohibited customer, activity, product, relationship type or jurisdiction under risk appetite or law.	Reject / Exit	Highest	Risk appetite reference, rejection/exit approval.
OVR-003	Synthetic/fictitious identity, forged/tampered ID, confirmed onboarding fraud.	Reject / Internal watchlist / Reporting review	Highest	Fraud evidence, IDSP result, investigation outcome.
OVR-004	Unable/refused to identify or verify UBO, controller or authorised signatory where required.	Reject or High pending remediation	High	UBO evidence gap, remediation record, Compliance decision.
OVR-005	Shell bank or FI permitting shell-bank use.	Reject / Exit	Highest	FI due diligence, respondent controls assessment.
OVR-006	Unlicensed regulated activity where a licence is required.	Reject / Prohibited	Highest	Licence review and legal/compliance assessment.
OVR-007	Anonymous, bearer-like or unverified product feature.	Prohibited	Highest	Product risk assessment and block evidence.
OVR-008	Foreign PEP, IO PEP, high-risk PEP nexus or close associate/family of material PEP.	High	High	PEP screening, SoW/SoF, senior approval.
OVR-009	Material adverse media involving ML/TF/PF, sanctions, fraud, corruption or serious predicate offence.	High or Reject	High	Adverse media review and mitigation rationale.
OVR-010	STR/SAR filed, confirmed suspicion or suspicion cannot be mitigated.	High	High	Restricted investigation record; tipping-off controls.
OVR-011	High-risk country exposure not prohibited, including high-risk residence/operation/SOF/corridor.	High	High	Country-risk library, EDD rationale and approval.
OVR-012	High-risk business, product or merchant activity within appetite.	High	High	Business/product risk assessment and EDD.
OVR-013	Failed or unresolved digital identity controls: liveness fail, biometric mismatch, deepfake/injection, device farm.	High or Reject	High	ID vendor result, fraud/compliance review.
OVR-014	Correspondent, nested, downstream or payable-through FI exposure within appetite.	High	High	FI EDD, downstream controls, senior approval.
OVR-015	Non-resident with strong evidence and no other High indicator.	Medium floor	Medium	Residency evidence, purpose, expected activity.
OVR-016	Moderate adverse media or domestic PEP with mitigants and no High indicator.	Medium floor	Medium	Disposition rationale and approval.
OVR-017	Unassessed country, medium-risk country, or multiple medium geography indicators.	Medium floor	Medium	Country risk task and remediation owner.
OVR-018	Document/manual digital verification resolved but not strong assurance.	Medium floor	Medium	Manual review evidence and approval.
OVR-019	New entity, limited operating history or newly launched merchant with plausible business model.	Medium floor	Medium	Operating evidence and follow-up review.
OVR-020	Explained VPN/proxy, moderate device anomaly or moderate transaction deviation.	Medium floor	Medium	Digital/fraud review and monitoring adjustment.

12.2 Manual Override Controls

- Manual uplifts may be applied by authorised Compliance users where emerging typology, weak evidence, unusual complexity or unresolved concern exists;
- Manual downgrades require Head of Compliance/MLRO or delegated senior Compliance approval, documented rationale, expiry/review date and evidence;
- Manual downgrades cannot reduce a prohibited outcome, hard stop or mandatory High/Medium floor below the required floor;
- All manual overrides must store before/after rating, reason code, narrative, approver, timestamp, evidence reference and next review date;
- Any deviation is otherwise managed through the formal dispensation and waiver process defined in the Policy (Section 9.10). Exceptions shall not be granted where prohibited by law, regulation, sanctions, TFS obligations or risk appetite.

13. Due Diligence, Approval, Monitoring and Review Outcomes

The final rating routes the customer to the required due diligence, approval authority, monitoring intensity and review frequency. Approval authorities operate within the Customer Acceptance Approval Matrix (Mal-PRC-ONB-01 Section 12) and the Policy's acceptance standard (Policy Section 11.2). The maximum review cycles in the outcome table below are set within, and are no less stringent than, the minimum periodic-review frequencies in Policy Section 14.1 (High at least annually; Medium at least every three years; Low at least every five years); where the two differ, the stricter (shorter) cycle applies.

Final outcome	CDD / EDD requirement	Approval authority	Review frequency	Monitoring intensity
Low	Standard CDD. SDD only where lawful, risk-supported, evidenced and not contradicted by facts.	Automated/Operations or business delegated approval where policy permits.	Maximum 36 months; earlier on trigger event.	Standard monitoring and baseline digital controls.
Medium	Standard CDD plus targeted additional information based on drivers.	Operations/Compliance approval depending on driver.	Maximum 24 months; earlier on event, DQ exception or activity deviation.	Moderate monitoring and selected enhanced scenarios.
High	EDD, SoF/SoW verification where relevant, adverse media review, enhanced sanctions/PEP checks and documented risk acceptance.	Head of Compliance/MLRO or delegated senior approver; senior management for PEP/correspondent where required.	Maximum 12 months; 6 months for very high-risk PEP/digital fraud/correspondent where risk appetite requires.	Enhanced monitoring, lower thresholds, periodic adverse media and restrictions where required.
Prohibited / Reject / Freeze / Exit	No relationship, or restricted legal/sanctions handling as required.	Compliance/MLRO/Legal escalation and governance approval.	N/A.	Block, freeze, reject, exit and report as required.

Note: The review-frequency maxima above are the model outcomes for this Methodology. Where the Policy or Mal-PRC-ONB-01 prescribes a stricter review cycle for a customer population, the stricter cycle prevails (Regulatory precedence, Section 2.3).

13.1 EDD Evidence Matrix

Risk driver	Minimum EDD evidence	System task fields
High geography	Purpose of relationship, reason for country nexus, SOF/SOW evidence, expected corridors, adverse media review.	EDD_GEO_RATIONALE, SOF_EVIDENCE, SOW_EVIDENCE, CORRIDOR_JUSTIFICATION.
High product / service	Product purpose, limit rationale, beneficiary/counterparty profile, product control confirmation.	PRODUCT_PURPOSE, LIMIT_APPROVAL, PRODUCT_CONTROL_CHECK.
PEP	PEP role, public function, country, SoW, expected activity, senior approval.	PEP_ROLE, PEP_SOW, SENIOR_APPROVAL.

Risk driver	Minimum EDD evidence	System task fields
Complex ownership	Ownership chart, UBO verification, legal purpose, source of wealth/funds.	OWNERSHIP_CHART, UBO_EVIDENCE, CONTROL_RATIONALE.
Digital ID exception	Exception reason, compensating verification, fraud review outcome, approval.	DIGITAL_EXCEPTION_REASON, COMPENSATING_CONTROL, FRAUD_REVIEW.
Merchant high risk	Website/app review, MCC validation, refund/chargeback controls, settlement review.	MERCHANT_DD, MCC_VALIDATION, SETTLEMENT_ACCOUNT_CHECK.
FI / regulated partner	Licence/regulator verification, AML questionnaire, sanctions controls, downstream exposure, audit/certification.	FI_DD_PACK, AML_QA, DOWNSTREAM_EXPOSURE, SANCTIONS_CONTROL_REVIEW.

14. Data Model, Source-System Mapping and Data Quality

14.1 Core Data Domains

Data domain	Source systems	Minimum implementation requirement
Customer master	Core banking, digital onboarding, CRM	Unique customer ID, segment, lifecycle status, residency, nationality, onboarding date, product holdings and relationship owner.
Related parties	KYC platform, onboarding forms, corporate registry integrations	UBOs, shareholders, directors, signatories, POAs, guardians, controllers, authorised users, merchant administrators and FI counterparties.
Screening	Sanctions/TFS, PEP/adverse media, internal watchlists	Screening status, match type, confidence, disposition, list source/version, date, reviewer and escalation outcome.
Digital identity	eKYC vendor, UAE Pass/EID validation, biometric/liveness engine	ID source, document authenticity, biometric score, liveness result, deepfake/injection signal and exception approval.
Device/channel	Mobile/web platform, fraud engine, device intelligence, WAF/SIEM	Device fingerprint, IP, GPS, SIM, VPN/TOR/proxy, emulator/root/jailbreak, velocity and ATO indicators.
Product	Core banking, product catalogue, payment platform, cards, merchant acquiring	Products requested/active, product risk version, features, limits, corridors, API access and control attributes.
Transactions and behaviour	Payment switch, cards, core banking, merchant acquiring, TM system	Actual turnover, count, velocity, deviation, corridors, counterparties, investigations, STR/SAR and fraud/mule cases.
Workflow/evidence	Case management, GRC, ticketing, document management, audit log	EDD tasks, approvals, rationale, evidence files, SLA, decision, reviewer, model version and history.

14.2 Minimum Field Catalogue

Field	Type	Requirement	Applies to	Rule
customer_id	String	Mandatory	All	Unique customer identifier from core or onboarding platform.
assessment_id	String	Mandatory	All	Unique risk assessment identifier.
model_version_id	String	Mandatory	All	Active approved model version used for calculation.
customer_segment	Enum	Mandatory	All	NP, LP, MER, FI, EMP or approved extension.
lifecycle_stage	Enum	Mandatory	All	New, Existing, Periodic Review, Event Review, Exit Review.
relationship_status	Enum	Mandatory	All	Applicant, Active, Suspended, Restricted, Exited, Rejected.
nationality_primary / secondary	ISO country	Mandatory / Conditional	NP	Primary and other nationality where declared/identified.
country_of_residence	ISO country	Mandatory	NP	Legal residence; missing country must not default to Low.

Field	Type	Requirement	Applies to	Rule
source_of_funds_country	ISO country	Mandatory	All	Primary country of recurring funds.
source_of_wealth_country	ISO country	Conditional	High risk / wealth	Country where wealth was generated.
country_of_incorporation / operation	ISO country	Mandatory	LP/MER/FI	Registration and primary operating jurisdiction.
ubos_countries	Array	Mandatory	LP/MER/FI	Nationality/residence of UBOs/controllers.
expected_corridors	Array	Mandatory	All	Expected transaction countries/corridors.
product_codes_requested / active	Array	Mandatory	All / Existing	Products requested or currently held.
expected_monthly_turnover	Decimal	Mandatory	All	Expected inflow/outflow or merchant turnover.
expected_txn_count	Integer	Mandatory	All	Expected monthly transaction count.
funding_method	Enum	Mandatory	All	Salary, business revenue, own account, third party, merchant settlement, loan proceeds, other.
digital_id_source	Enum	Mandatory digital	NP / signatories	UAE Pass, Emirates ID, approved IDSP, document upload, assisted video, branch.
document_auth_result	Enum	Mandatory digital	NP / signatories	Pass, fail, manual review, not applicable.
biometric_match_score	Decimal / Enum	Mandatory digital	NP / signatories	Vendor score or pass/fail; threshold stored by model version.
liveness_result	Enum	Mandatory digital	NP / signatories	Pass, fail, inconclusive, not performed.
device_integrity_status	Enum	Mandatory digital	Digital	Normal, rooted/jailbroken, emulator, tampered, unknown.
vpn_tor_proxy_flag	Boolean	Mandatory digital	Digital	True where VPN/TOR/proxy/anonymiser detected.
screening_status	Enum	Mandatory	All	Clear, potential match, true match, false positive, pending.
pep_status	Enum	Mandatory	NP / related parties	None, domestic PEP, foreign PEP, IO PEP, associate/family.
adverse_media_status	Enum	Mandatory	All	None, low, medium, high, true material.
sar_str_history	Enum/Integer	Mandatory existing	Existing	Number/status of prior SAR/STR or internal suspicion cases.
final_risk_rating	Enum	Mandatory	All	Low, Medium, High, Prohibited/Reject.

14.3 Data-Quality Treatment Rules

Issue	Treatment	Examples
Critical KYC identity data missing	Block onboarding / block final assessment.	Identity, legal existence, UBO, nationality/residence, licence, authorised signatory.
Country data missing or unmapped	Treat at least Medium; High if other risk indicators exist; create remediation task.	Country must not default to UAE or Low.
Occupation/industry unmapped	Treat at least Medium pending Compliance mapping; High if high-risk indicators exist.	Unmapped self-employed industry or MCC.
Product unmapped	Treat at least Medium and route to product risk owner; prohibit activation of unapproved product.	New product, virtual account feature, API access.
Activity thresholds missing	Block final approval or apply conservative High to activity until expected activity is captured.	Do not use zero where data is missing.
Screening pending	Hold activation until cleared or resolved.	Sanctions, PEP, adverse media and related-party screening.

Issue	Treatment	Examples
Manual downgrade requested	Require Compliance approval, rationale, evidence and expiry/review date.	System retains mathematical score, original rating, requested rating and approver.

15. Calculation Sequence, Workflow and SLAs

15.1 Calculation Sequence

Step	System action	Control
1	Validate segment, lifecycle stage and mandatory data.	If critical blockers exist, assessment status = DATA_PENDING or REJECTED.
2	Run or retrieve screening status for customer and material related parties.	Confirmed sanctions/TFS true match returns hard stop OVR-001. Pending match creates SCREENING_PENDING or SANCTIONS_HOLD.
3	Load active approved model version and reference libraries.	Only one active approved parameter set per effective date.
4	Score parameters using libraries/rules and generate reason codes.	Every non-Low parameter requires a reason code and source field.
5	Calculate factor scores using configured methods.	Weights must sum to 1.000000 within tolerance; factor scores must remain 1.0000–3.0000.
6	Calculate raw composite score and mathematical rating.	Use unrounded raw score and authoritative bands.
7	Evaluate overrides and floors in priority order.	Store all triggered rules; apply most restrictive outcome.
8	Derive CDD/EDD, approval, monitoring and review outcomes.	High-risk approval before activation unless restricted handling is approved.
9	Persist assessment snapshot, evidence, factor/parameter scores and audit log.	Completed assessments immutable; corrections create a new version.
10	Publish final approved rating and restrictions to downstream systems.	Core banking, TM, screening, CRM, product systems and MI repository.

15.2 Workflow States

State	Description	Allowed next states
DRAFT	Assessment started but mandatory data incomplete.	DATA_PENDING, SCREENING_PENDING, CALCULATION_READY, ABANDONED.
DATA_PENDING	Mandatory fields/evidence missing or failed validation.	CALCULATION_READY, REJECTED, DRAFT.
SCREENING_PENDING	Mandatory screening not completed or potential match unresolved.	CALCULATION_READY, SANCTIONS_HOLD, REJECTED.
CALCULATION_READY	Inputs validated and risk engine can calculate.	CALCULATED, ERROR.
CALCULATED	Composite rating and override result generated.	EDD_REQUIRED, APPROVAL_REQUIRED, AUTO_APPROVED, REJECTED.
EDD_REQUIRED	EDD checklist/evidence required.	APPROVAL_REQUIRED, REJECTED, DATA_PENDING.
APPROVAL_REQUIRED	Approval required based on rating/floor/override.	APPROVED, REJECTED, MORE_INFO_REQUIRED.
SANCTIONS_HOLD	Potential/true sanctions match handling in progress.	REJECTED, EXITED, APPROVED only if false positive cleared.
RESTRICTED	Customer active with restricted products/channels/limits.	ACTIVE, EXITED, PERIODIC_REVIEW.

State	Description	Allowed next states
APPROVED / ACTIVE	Relationship may activate/continue subject to controls.	PERIODIC_REVIEW, EVENT_REVIEW, RESTRICTED, EXITED.
REJECTED / EXITED	Customer not onboarded or relationship exited.	Closed state except approved appeal/remediation workflow.

15.3 Event Triggers and SLA Matrix

Trigger event	Severity	Source	Required action	SLA
Sanctions/TFS potential or true match	Critical	Screening system	Hold/freeze/escalate; prohibit if true match.	Immediate / same day per sanctions procedure.
FATF/country library update	High	Compliance library	Batch rescore affected customers/related parties.	Within approved change window; critical updates immediate.
New product, feature or limit increase	Medium/High	Product/core banking	Pre-activation rescore.	Before activation.
Change in nationality/residence/address/SOF	Medium/High	Customer servicing/KYC	Event-driven review and rescore.	Within 5 business days or before high-risk transaction.
UBO/shareholder/director/signatory change	High	Corporate KYC	CDD refresh, screening and rescore.	Before acceptance or within policy SLA.
PEP/adverse media true match	High	Screening/adverse media	EDD task and rescore.	Within policy SLA.
STR/SAR or confirmed suspicion	High	AML investigations	High floor, enhanced monitoring and exit/risk acceptance review.	Immediate after decision.
Device/account takeover event	High	Fraud/cyber/device monitoring	Restrict high-risk activity; rescore channel/behaviour.	Immediate for transaction risk.
Transaction behaviour threshold breach	Medium/High	TM/payment systems	Create review case; High triggers immediate rescore.	Daily or near-real-time based on severity.
Periodic review due	Scheduled	KYC system	Refresh data and recalculate.	As per rating cycle.

16. Record Keeping, Audit Trail and Explainability

The Bank maintains comprehensive, accurate, secure and readily retrievable CRA records to support its AML/CFT/CPF compliance framework and to respond promptly to requests from the CBUAE, FIU, law enforcement, judicial and other competent authorities. All CRA records are retained for a minimum of five (5) years, or longer where required, in accordance with Policy Section 21 and Mal-PRC-ONB-01 Section 13, and are protected from unauthorised access, alteration, deletion, loss or destruction.

Audit requirement	Implementation detail
Point-in-time reproducibility	Store raw input values, transformed values, model version, country/product/library versions, factor/parameter scores, overrides, approvals and evidence references.
User/system actor tracking	Record actor ID for data entry, system-triggered calculation, review, approval, override, rejection and publication.

Audit requirement	Implementation detail
Immutable completed assessment	Completed assessments must not be overwritten. Corrections require a new assessment version linked to the prior assessment.
Evidence linkage	Each Medium/High factor, override and EDD driver must link to supporting evidence or rationale.
Override log	All overrides store rule/manual flag, reason code, narrative, approver, timestamp, before/after rating and expiry/review date.
Data lineage	Each critical field stores source system, extraction timestamp, transformation logic and reconciliation status.
Explainability output	UI and MI show raw score, mathematical rating, final rating, top drivers, floors/overrides, required controls and next review date.

17. Model Governance, Validation and Change Control

17.1 Production Validation Gates

Gate	Objective	Minimum exit criteria	Approver
Gate 0 — Model freeze	Confirm one approved methodology and parameter set.	Methodology and parameter set approved and frozen; single active configuration confirmed; model_version_id issued. Validation design findings F-001 to F-004 and F-006 to F-009 closed for design (see Validation gap closure register).	Model Governance / FCC / MLRO.
Gate 1 — Data readiness	Confirm source-system mapping and data quality.	Source-to-target matrix complete; DQ test passed; lineage and exception register approved.	Data Governance / Technology / FCC.
Gate 2 — Vendor readiness	Confirm vendor outputs are fit for intended use.	Vendor due diligence, performance metrics, integration mapping and monitoring cadence approved.	Vendor Governance / Fraud / FCC.
Gate 3 — SIT exit	Confirm rule engine, APIs, batch jobs, audit and security work.	100% mandatory scripts executed; zero open Critical/High defects.	Technology QA / Compliance Technology.
Gate 4 — UAT exit	Confirm business users can operate the model end-to-end.	Business scenarios passed; user acceptance signed; training complete.	Operations / FCC / Fraud / Sanctions.
Gate 5 — Calibration, parallel run and back-testing	Confirm outcome reasonableness and operational impact.	Calibration, movement matrix, false-negative review and operational impact approved.	Model Validation / FCC / Operations.
Gate 6 — Production approval	Approve go-live.	All Critical/High findings closed; Medium residuals accepted; monitoring plan active.	Senior Governance Forum.

17.2 Quantitative Calibration Requirements

- Use a representative 12–24 month population covering customer segments, products, geographies, channels and lifecycle stages;
- Produce rating distribution, threshold sensitivity, outcome-rate-by-band, override impact, false-negative analysis, stability/drift metrics and segment/product/geography concentration analysis;
- Demonstrate directional risk sensitivity: High-rated and overridden customers should show higher adverse-outcome incidence than Medium and Low populations, subject to explanation for portfolio composition;
- Document any threshold or weight tuning with before/after movement, risk rationale, operational impact and governance approval;
- Link calibration approval to the frozen model_version_id.

17.3 Change Control

Change type	Examples	Approval	Testing requirement
Minor configuration	Typo, display label, non-risk workflow text.	Compliance Technology + Owner.	Regression smoke test.
Reference library update	Country, product, occupation, industry or MCC score update.	Financial Crime Compliance.	Impact analysis and affected-customer rescore plan.
Threshold / weight change	Rating bands, factor weights or parameter weights.	Model Governance Committee / Risk Committee.	SIT, UAT, parallel run, validation sign-off.
New product/channel	New payment product, API feature, wallet, merchant channel.	Product Governance + Compliance + Risk.	Product financial-crime risk assessment and scenario testing.
Vendor/model change	Digital ID vendor, screening algorithm, device risk model.	Vendor Governance + Compliance + Technology Risk.	Assurance review, performance test and fallback plan.
Regulatory change	CBUAE notice, sanctions requirement, FATF update.	Compliance + Legal + Risk Committee as required.	Gap assessment and implementation plan.

17.4 Model Version Control and Configuration Baseline

The canonical model version for the model package is CRAM-CBUAE-2026-05-FREEZE-01. This Methodology, the End-to-End Customer Lifecycle and Onboarding Operating Model, and the dataset workbook must all reference this exact identifier. Point-in-time reproducibility requires that the model version, the reference-library versions and the effective date resolve to this frozen configuration.

Configuration area	Baseline rule	Owner / gate
Model version identity	CRAM-CBUAE-2026-05-FREEZE-01 is the single canonical model_version_id, applied consistently in the document control table, the closing control statement, the page footer, the operating model and the workbook data dictionary.	Model Owner / Gate 0
Factor and parameter structure	Scoring is two-layer: parameters (Section 6) roll up into the composite factors, which combine at the segment and lifecycle weights (Section 5.1). The workbook factor/parameter map records every parameter-to-factor assignment. All weight sets sum to 100%.	Model Owner / Gate 0
FI systems-and-controls counting	The FI systems-and-controls factor (Section 6.3.2) enters the composite once at its Section 5.1 factor weight. FI control parameters roll up into this factor and are not added to the composite independently.	Model Owner / Gate 0
Geography scale	Country risk is expressed through the separate Section 7.1 pillars; each pillar carries a single coherent value domain. The composite country risk level is derived from the weighted pillars.	Model Owner / Gate 0
Reference libraries	Country, occupation/industry/MCC and product reference libraries are maintained as controlled, versioned, executable data with maker-checker and effective-date control; every scoring lookup resolves against a named, versioned table.	Data Governance / Gate 1
Vendor-dependent thresholds	Identity-proofing, document-authenticity, biometric/liveness, device-integrity and screening-match thresholds are model-version-bound parameters confirmed under vendor validation; conservative floors apply to weak or inconclusive outcomes until thresholds are approved.	Vendor Governance / Gate 2
Calibration and back-testing	Outcome reasonableness, rating distribution, threshold sensitivity, override impact, false-negative analysis and stability/drift are confirmed on a representative population and tied to the frozen model_version_id before production approval.	Model Validation / Gate 5

The non-dilution principle governs every item above: no configuration may weaken a hard stop, a mandatory floor or an override, and strong controls may never reduce inherent risk below the level implied by inherent risk drivers.

18. MI, KRIs and Ongoing Monitoring

KRI / MI	Frequency	Indicator	Owner
Portfolio distribution by risk rating	Monthly	High-risk concentration above risk appetite threshold or sudden rating shifts.	Compliance / Risk Committee.
High-risk onboarding approval SLA	Monthly	EDD/approval overdue beyond SLA.	Compliance Operations.
Override volume and direction	Monthly	High manual override volume, downgrades or repeated reason codes.	Model Governance.
Digital ID exception rate	Monthly	Exception rate above tolerance, vendor drift or failed liveness increase.	Digital Onboarding / Fraud / FCC.
Screening potential/true match trends	Monthly	Increase in true matches, unresolved potential matches or list-update impact.	Sanctions Compliance.
Geography exposure by country/corridor	Monthly	Growth in high-risk country exposure or new corridor concentrations.	FCC / Product.
Product risk exposure	Monthly	Increase in high-risk product customers, limits or volume.	Product / FCC.
TM alerts and STR/SAR by risk rating	Monthly	Low-rated customers producing disproportionate suspicious activity.	AML Investigations / Model Validation.
Data-quality exceptions	Monthly	Mandatory field gaps, stale CDD, missing related parties, unresolved mappings.	Data Governance.
Model performance / back-testing	Quarterly	Rating not predictive of alerts/STRs or high false-negative indicators.	Model Validation.

19. Final Methodology Control Statement

This Methodology is the single authoritative Customer Risk Assessment Methodology for model version CRAM-CBUAE-2026-05-FREEZE-01. No alternative threshold set, FI factor taxonomy, unweighted FI parameter library, undefined Medium floor, uncontrolled control-effectiveness reduction or obsolete boundary test may be used in production configuration, validation evidence, user training, MI reporting or governance packs. Any change to the threshold, factor taxonomy, weights, floors, overrides or product/control treatment requires formal change control, impact assessment, validation and governance approval before deployment (Section 17).

20. Appendices

The following appendices form an integral part of this Methodology and shall be read in conjunction with the AML/CFT/CPF Compliance Policy and Mal-PRC-ONB-01. Where any appendix conflicts with the Policy, the Policy prevails.

Appendix 1 — UAT, SIT and Validation Test Pack

Test ID	Scenario	Expected outcome
TC-001	Low-risk resident NP, strong digital ID, low-risk product, no alerts.	Low rating; standard CDD; activation permitted; 36-month maximum review.
TC-002	Raw score exactly 1.5000 with no override.	Low mathematical rating; threshold decision uses unrounded raw score.
TC-003	Raw score exactly 1.5001 with no override.	Medium mathematical rating; CDD plus targeted controls.
TC-004	Raw score exactly 2.1500 with no override.	Medium mathematical rating.
TC-005	Raw score exactly 2.1501 with no override.	High mathematical rating; EDD and approval.
TC-006	Foreign PEP with otherwise Low profile.	High final rating due to non-dilutable PEP floor.
TC-007	Confirmed sanctions/TFS true match.	Hard stop/freeze/escalate; no onboarding.
TC-008	Legal entity refuses to disclose UBO.	Reject/prohibited or High pending remediation; no composite-score dilution.

Test ID	Scenario	Expected outcome
TC-009	SME with complex but verified ownership and medium-risk product.	Medium or High depending on complexity; EDD where required.
TC-010	Merchant acquiring request for high-risk MCC and cross-border settlement.	High rating; merchant EDD and product approval before activation.
TC-011	Resident customer using document upload only with liveness not performed.	High or onboarding hold due to weak digital assurance.
TC-012	Device farm detected across multiple applications.	High/fraud review; linked applications held.
TC-013	Existing Low customer suddenly receives 5x expected turnover with rapid pass-through.	Event-driven rescore; Medium/High; TM investigation.
TC-014	Country library changes a country from Medium to High.	Batch rescore affected customers; workflow tasks for newly High customers.
TC-015	New product added: international transfer.	Pre-activation rescore; product/geography monitoring rules activated.
TC-016	Manual downgrade attempted below mandatory High floor.	System blocks or requires policy-authorised exception; cannot go below floor.
TC-017	False positive sanctions match cleared.	Screening status updated; rating recalculated; audit trail retained.
TC-018	FI partner with downstream high-risk corridor exposure.	High; FI EDD, sanctions controls review and senior approval.
TC-019	Unassessed country in source of wealth field.	At least Medium pending assessment; High if combined with other risk indicators.
TC-020	Product control weakness identified in payment API.	Risk uplift/remediation; no control-based reduction below inherent score.
TC-021	STR/SAR filed on existing customer.	High floor, enhanced monitoring and exit/risk acceptance assessment.
TC-022	Periodic review overdue for High customer.	Workflow breach, escalation and possible restriction according to policy.

Appendix 2 — Implementation Checklist

#	Checklist item	Owner	Evidence
1	Approve this Methodology and issue the frozen model_version_id.	Model Governance / MLRO	Approval minutes and model version record.
2	Configure authoritative thresholds and active factor weights.	Technology / FCC	Configuration export and deployment validation.
3	Configure segment and parameter libraries, including FI parameter weights.	FCC / Technology	Parameter table extract.
4	Configure hard stops, High floors and Medium floors with priorities.	FCC / Technology	Override rule table and test evidence.
5	Map all mandatory fields to source systems and owners.	Data Governance	Source-to-target matrix and lineage map.
6	Complete data-quality testing and remediation/exception register.	Data Governance / Operations	DQ report and exception approvals.
7	Complete vendor validation for screening, digital ID, device and fraud services.	Vendor Governance / FCC / Fraud	Vendor due diligence and performance reports.
8	Complete SIT and close all Critical/High defects.	Technology QA	SIT exit report.
9	Complete UAT with Compliance, Operations, Fraud and Sanctions users.	Business / FCC	UAT sign-off and evidence.
10	Complete calibration, parallel run and back-testing.	Model Validation / FCC	Calibration pack, movement matrix, back-test report.
11	Complete MI/KRI dashboards and post-go-live monitoring plan.	Compliance MI	Dashboard evidence and reporting calendar.

#	Checklist item	Owner	Evidence
12	Obtain production go-live approval.	Senior Governance Forum	Gate 6 approval and residual-risk acceptance.

Appendix 3 — Sample Output Payloads

This appendix provides three illustrative system output payloads to support technology configuration, API design, SIT/UAT execution, model validation, auditability and regulatory review. The payloads show how the model stores mathematical score, final rating, thresholds, factor contributions, overrides, data-quality status, screening outcomes, CDD/EDD requirements, approvals, monitoring profile and point-in-time audit metadata. Payload field names are illustrative and must be harmonised with the Bank enterprise data dictionary before production deployment.

The three scenarios cover: (1) Low-risk customer with no override; (2) High-risk customer where an override or floor uplifts the rating; and (3) Prohibited / hard-stop case where sanctions or TFS controls block onboarding or continuation. All payloads are bound to the authoritative model version CRAM-CBUAE-2026-05-FREEZE-01 and the authoritative rating bands in Section 3.1.

A3.1 Low-Risk Customer — No Override

Illustrative outcome where all mandatory data is complete, screening is clear, digital identity checks pass, factor scores remain Low, and no override or risk floor applies.

```
{
  "assessment_id": "CRA-20260524-000001",
  "score_run_id": "CRR-20260524-000001",
  "customer_id": "CUST-LR-0001",
  "customer_segment": "NATURAL_PERSON",
  "lifecycle_stage": "NEW_ONBOARDING",
  "relationship_status": "APPLICANT",
  "model_version_id": "CRAM-CBUAE-2026-05-FREEZE-01",
  "assessment_timestamp": "2026-05-24T09:15:00+04:00",
  "rating_thresholds": {
    "low": "1.0000-1.5000",
    "medium": "1.5001-2.1500",
    "high": "2.1501-3.0000"
  },
  "mathematical_score_raw": 1.2000,
  "mathematical_score_display": 1.20,
  "mathematical_rating": "Low",
  "final_rating": "Low",
  "factor_scores": [
    {
      "factor_code": "NP_PROFILE",
      "factor_name": "Customer Profile Risk",
      "factor_rating": 1,
      "factor_weight": 0.25,
      "weighted_contribution": 0.2500,
      "reason_codes": ["SALARIED_VERIFIED_EMPLOYER", "NO_PEP_EXPOSURE", "CLEAR_SOURCE_OF_FUNDS"]
    },
    {
      "factor_code": "NP_GEO",
      "factor_name": "Geography and Sanctions Nexus Risk",
      "factor_rating": 1,
      "factor_weight": 0.20,
      "weighted_contribution": 0.2000,
      "reason_codes": ["LOW_RISK_RESIDENCE_COUNTRY", "LOW_RISK_NATIONALITY"]
    },
    {
      "factor_code": "NP_PRODUCT",
      "factor_name": "Product and Service Risk",
      "factor_rating": 1,
      "factor_weight": 0.20,
      "weighted_contribution": 0.2000,
      "reason_codes": ["SALARY_ACCOUNT_LOW_LIMITS", "NO_CROSS_BORDER_FEATURE"]
    },
    {
      "factor_code": "NP_CHANNEL",
      "factor_name": "Digital Onboarding and Channel Assurance",
      "factor_rating": 1,
      "factor_weight": 0.25,
      "weighted_contribution": 0.2500,
      "reason_codes": ["UAE_PASS_VERIFIED", "BIOMETRIC_LIVENESS_PASSED", "DEVICE_INTEGRITY_NORMAL"]
    }
  ]
}
```

```

"factor_code": "NP_ACTIVITY",
"factor_name": "Expected Activity and Transaction Behaviour",
"factor_rating": 1,
"factor_weight": 0.10,
"weighted_contribution": 0.1000,
"reason_codes": ["LOW_EXPECTED_MONTHLY_TURNOVER", "SALARY_FUNDING_ONLY"]
},
],
"overrides_and_floors": [],
"data_quality_status": "Complete",
"screening_status": {
"sanctions_tfs": "Clear",
"pep": "No Match",
"adverse_media": "No Material Match",
"internal_watchlist": "Clear",
"screening_reference": "SCR-20260524-000001"
},
"required_due_diligence": "Standard CDD",
"edd_requirements": [],
"approval_route": {
"required_approval": "Automated or Operations Approval",
"approval_status": "Approved",
"approver_role": "Operations Maker/Checker or Automated Rule Engine"
},
"monitoring_profile": {
"transaction_monitoring_intensity": "Standard",
"screening_frequency": "Ongoing list screening and event-driven review",
"review_cycle_months": 36,
"next_review_date": "2029-05-24"
},
"audit_metadata": {
"created_by": "CRAM_RULE_ENGINE",
"source_systems": [
"Digital Onboarding",
"Core Banking",
"Screening System",
"Product Catalogue"
],
"country_library_version": "COUNTRY-RISK-2026-05",
"product_library_version": "PRODUCT-RISK-2026-05",
"override_library_version": "OVERRIDE-2026-05",
"input_snapshot_hash": "sha256:low-risk-input-snapshot",
"output_snapshot_hash": "sha256:low-risk-output-snapshot"
}
}

```

A3.2 High-Risk Customer — Override / Risk Floor Applied

Illustrative outcome where the mathematical score is Medium, but the final rating is uplifted to High because high-risk merchant activity and card-not-present exposure trigger mandatory High floors.

```

{
"assessment_id": "CRA-20260524-000002",
"score_run_id": "CRR-20260524-000002",
"customer_id": "CUST-HR-0002",
"customer_segment": "LEGAL_PERSON_MERCHANT",
"lifecycle_stage": "NEW_ONBOARDING",
"relationship_status": "APPLICANT",
"model_version_id": "CRAM-CBUAE-2026-05-FREEZE-01",
"assessment_timestamp": "2026-05-24T11:40:00+04:00",
"rating_thresholds": {
"low": "1.0000-1.5000",
"medium": "1.5001-2.1500",
"high": "2.1501-3.0000"
},
"mathematical_score_raw": 2.0500,
"mathematical_score_display": 2.05,
"mathematical_rating": "Medium",
"final_rating": "High",
"factor_scores": [
{
"factor_code": "LP_PROFILE",
"factor_name": "Customer Profile Risk",
"factor_rating": 2,
"factor_weight": 0.25,
"weighted_contribution": 0.5000,
"reason_codes": ["MULTI_LAYER_OWNERSHIP_VERIFIED", "NEW_ENTITY_LIMITED_OPERATING_HISTORY"]
},
{

```

```

"factor_code": "LP_GEO",
"factor_name": "Geography and Sanctions Nexus Risk",
"factor_rating": 2,
"factor_weight": 0.20,
"weighted_contribution": 0.4000,
"reason_codes": ["MEDIUM_RISK_OPERATING_COUNTRY", "MIXED_COUNTERPARTY_CORRIDORS"]
},
{
"factor_code": "LP_PRODUCT",
"factor_name": "Product and Service Risk",
"factor_rating": 3,
"factor_weight": 0.20,
"weighted_contribution": 0.6000,
"reason_codes": ["MERCHANT_ACQUIRING", "CARD_NOT_PRESENT", "HIGH_RISK_MCC"]
},
{
"factor_code": "LP_CHANNEL",
"factor_name": "Digital Onboarding and Channel Assurance",
"factor_rating": 1,
"factor_weight": 0.20,
"weighted_contribution": 0.2000,
"reason_codes": ["VERIFIED_DIGITAL_ID", "SIGNATORY_LIVENESS_PASSED"]
},
{
"factor_code": "LP_ACTIVITY",
"factor_name": "Expected Activity and Transaction Behaviour",
"factor_rating": 2,
"factor_weight": 0.15,
"weighted_contribution": 0.3000,
"reason_codes": ["MEDIUM_EXPECTED_MONTHLY_TURNOVER", "CROSS_BORDER_SETTLEMENT_EXPECTED"]
}
],
"overrides_and_floors": [
{
"rule_id": "OVR-010",
"rule_name": "High-Risk Business Activity Within Risk Appetite",
"trigger_condition": "Merchant category classified as high-risk but not prohibited",
"outcome": "High",
"priority": "High",
"evidence_required": true,
"evidence_reference": "EDD-MER-20260524-000002"
},
{
"rule_id": "OVR-014",
"rule_name": "High-Risk Merchant / Card-Not-Present Exposure",
"trigger_condition": "Merchant acquiring requested for high-risk MCC with cross-border settlement exposure",
"outcome": "High",
"priority": "High",
"evidence_required": true,
"evidence_reference": "MERCHANT-RISK-20260524-000002"
}
],
"data_quality_status": "Complete",
"screening_status": {
"sanctions_tfs": "Clear",
"pep": "No Match",
"adverse_media": "No Material Match",
"internal_watchlist": "Clear",
"screening_reference": "SCR-20260524-000002"
},
"required_due_diligence": "EDD",
"edd_requirements": [
"Merchant due diligence",
"Business model rationale",
"Website and application review",
"MCC validation",
"UBO verification",
"Source of funds evidence",
"Settlement account review",
"Chargeback and refund controls assessment",
"Expected cross-border corridor rationale"
],
"approval_route": {
"required_approval": "Senior Compliance Approval",
"approval_status": "Pending",
"approver_role": "MLRO / Head of Compliance or Delegated Senior Compliance Approver"
},
"monitoring_profile": {
"transaction_monitoring_intensity": "Enhanced",
"screening_frequency": "Ongoing, periodic, and event-driven",
"merchant_monitoring": "Enhanced chargeback, refund, settlement, and MCC monitoring",
"review_cycle_months": 12,

```

```

"next_review_date": "2027-05-24"
},
"audit_metadata": {
"created_by": "CRAM_RULE_ENGINE",
"source_systems": [
"Digital Onboarding",
"Core Banking",
"Screening System",
"Product Catalogue",
"Merchant Acquiring Platform",
"Case Management"
],
"country_library_version": "COUNTRY-RISK-2026-05",
"product_library_version": "PRODUCT-RISK-2026-05",
"merchant_library_version": "MERCHANT-RISK-2026-05",
"override_library_version": "OVERRIDE-2026-05",
"input_snapshot_hash": "sha256:high-risk-input-snapshot",
"output_snapshot_hash": "sha256:high-risk-output-snapshot"
}
}

```

A3.3 Prohibited / Hard-Stop Case — Sanctions or TFS True Match

Illustrative outcome where a confirmed sanctions or TFS true match prevents ordinary scoring completion and requires hard-stop handling, restriction, escalation, freeze/reporting assessment, and restricted-access evidence.

```

{
"assessment_id": "CRA-20260524-000003",
"score_run_id": "CRR-20260524-000003",
"customer_id": "CUST-PHB-0003",
"customer_segment": "NATURAL_PERSON",
"lifecycle_stage": "NEW_ONBOARDING",
"relationship_status": "APPLICANT",
"model_version_id": "CRAM-CBUAE-2026-05-FREEZE-01",
"assessment_timestamp": "2026-05-24T14:05:00+04:00",
"rating_thresholds": {
"low": "1.0000-1.5000",
"medium": "1.5001-2.1500",
"high": "2.1501-3.0000"
},
"mathematical_score_raw": null,
"mathematical_score_display": null,
"mathematical_rating": "Not Calculated",
"final_rating": "Prohibited / Hard Stop",
"factor_scores": [],
"overrides_and_floors": [
{
"rule_id": "OVR-001",
"rule_name": "Confirmed Sanctions / TFS Match",
"trigger_condition": "Confirmed true match against applicable UN / UAE TFS or sanctions list",
"outcome": "Hard Stop / Freeze / Report",
"priority": "Highest",
"evidence_required": true,
"evidence_reference": "SANCTIONS-CASE-20260524-000003"
}
],
"hard_stop_details": {
"hard_stop_triggered": true,
"hard_stop_reason": "Confirmed sanctions / TFS true match",
"onboarding_allowed": false,
"account_activation_allowed": false,
"transaction_allowed": false,
"freeze_required": true,
"regulatory_escalation_required": true,
"case_restriction_required": true
},
"data_quality_status": "Screening True Match - Hard Stop",
"screening_status": {
"sanctions_tfs": "True Match",
"pep": "Not Assessed Due To Hard Stop",
"adverse_media": "Not Assessed Due To Hard Stop",
"internal_watchlist": "Escalated",
"screening_reference": "SCR-20260524-000003",
"sanctions_case_reference": "SANCTIONS-CASE-20260524-000003"
},
"required_due_diligence": "No onboarding / sanctions escalation",
"edd_requirements": [
"Sanctions true-match confirmation evidence",
"Screening disposition record",

```

```
"Legal and Compliance escalation",
"Freeze/reporting assessment",
"Regulatory reporting evidence where applicable",
"Restricted-access case notes to prevent tipping-off"
],
"approval_route": {
"required_approval": "MLRO / Sanctions Officer / Legal Escalation",
"approval_status": "Escalated",
"approver_role": "MLRO, Sanctions Officer, and Legal where required"
},
"monitoring_profile": {
"transaction_monitoring_intensity": "Blocked / Restricted",
"screening_frequency": "Immediate sanctions escalation and ongoing restriction",
"review_cycle_months": null,
"next_review_date": null
},
"audit_metadata": {
"created_by": "CRAM_RULE_ENGINE",
"source_systems": [
"Digital Onboarding",
"Screening System",
"Sanctions Case Management",
"Core Banking"
],
"country_library_version": "COUNTRY-RISK-2026-05",
"product_library_version": "PRODUCT-RISK-2026-05",
"override_library_version": "OVERRIDE-2026-05",
"screening_list_version": "SANCTIONS-LIST-2026-05-24",
"input_snapshot_hash": "sha256:hard-stop-input-snapshot",
"output_snapshot_hash": "sha256:hard-stop-output-snapshot"
}
}
```